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Homework[0]: LaTeX & GitHub Demonstration

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1. LINKS TO CODE

Here is a link to a file for this project: https://github.com/HaikunXue/OU-ASTR-5900/blob/main/HW0_file.rm

It is a list of food I have recently, I was trying to be descriptive but was too busy with Stat Mech HW.

2. MAIN TASK

How do we define a sentence? Is it a line of text, with a period? Or does question mark or exclamation mark also ends a sentence! We will find out. `print("This is a sentence.")`

2.1. *Reference to Figure*

Figure 1 is my figure for this homework, as it is the astronomy image of the day from NASA for today, 1/30/2026. It is a long exposure image of NGC 1333, near Perseus. We see jets and glowing gas cloud, and some potential star forming regions. Is this what our galaxy looks like in the past?

2.2. *Equation Flex*

$$du = \frac{\partial u}{\partial S} dS + \frac{\partial u}{\partial v} dv = Tds - Pdv \tag{1}$$

Figure 1. Here is a sample figure, the astronomy image of the day of a nebula with distinct blue and red region, and clear dust regions around the AGN

This is the Euler's equation from Thermodynamics, might be useful in astronomical simulations of star formation gas clouds.

3. QUESTION 2

There is only one question, but more to come! Lots of blank space that could be used better for next assignment.